

Annexure A

Analytical Problem Solving

- 1.** Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and a 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into the 4-gallon jug? Explicit assumptions: a jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another, and there are no other measuring devices available.

- 2.** Find out how the Mars Rover analyses and explores samples on Mars by answering the following questions: (i) What are the percepts for this agent? (ii) Characterize the operating environment. (iii) What are the actions the agent can take? (iv) How can one evaluate the performance of the agent? (v) What sort of agent architecture is most suitable for this agent, and why?

- 3.** Explore the search space using search strategies and formulate the problem components for the 8-Queens problem: place 8 queens on a chessboard such that none of the queens attack any of the others. A configuration of 8 queens on the board is shown in the assignment figure, but it does not represent a solution since the queen in the first column is on the same diagonal as the queen in the last column.

- 4.** A customer wants to travel from one location to another using the OLA Cab booking mobile application. The customer can select any cab type - mini, micro, sedan, shared, prime, and so on - based on comfort, to reach the destination. Identify what type of problem-solving agent can be used, and write the pseudocode for the above problem.

- 5.** A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm. The delivery network is represented as a weighted graph (see attached figure).